

Complete YOLO/SPPF neural ablation matrix across streams, heads, capacities, and seeds

A. Held-out frame macro-F1: LSTM

Full	0.747 ±0.006	0.731 ±0.003	0.725 ±0.004
No group visual	0.734 ±0.007	0.728 ±0.004	0.728 ±0.003
Pose geometry	0.711 ±0.008	0.711 ±0.005	0.712 ±0.009
No pairwise geometry	0.687 ±0.017	0.698 ±0.003	0.696 ±0.010
Animal visual	0.638 ±0.023	0.633 ±0.018	0.636 ±0.027
Visual only	0.672 ±0.003	0.668 ±0.004	0.675 ±0.003
	256	512	896

B. Held-out frame macro-F1: attention

Full	0.752 ±0.006	0.738 ±0.005	0.743 ±0.003
No group visual	0.738 ±0.007	0.739 ±0.005	0.733 ±0.004
Pose geometry	0.724 ±0.007	0.727 ±0.001	0.728 ±0.004
No pairwise geometry	0.696 ±0.005	0.689 ±0.012	0.697 ±0.012
Animal visual	0.680 ±0.003	0.676 ±0.009	0.675 ±0.008
Visual only	0.687 ±0.003	0.680 ±0.007	0.687 ±0.004
	256	512	896

C. Held-out bout macro-F1 at TIoU 0.25: LSTM

Full	0.672 ±0.002	0.659 ±0.011	0.643 ±0.006
No group visual	0.659 ±0.007	0.641 ±0.006	0.641 ±0.001
Pose geometry	0.641 ±0.006	0.642 ±0.009	0.640 ±0.002
No pairwise geometry	0.616 ±0.013	0.632 ±0.010	0.627 ±0.010
Animal visual	0.557 ±0.025	0.550 ±0.012	0.545 ±0.023
Visual only	0.593 ±0.006	0.581 ±0.005	0.595 ±0.005
	256	512	896

D. Held-out bout macro-F1 at TIoU 0.25: attention

Full	0.680 ±0.011	0.664 ±0.010	0.671 ±0.006
No group visual	0.665 ±0.007	0.666 ±0.010	0.657 ±0.007
Pose geometry	0.663 ±0.006	0.657 ±0.007	0.662 ±0.002
No pairwise geometry	0.610 ±0.006	0.606 ±0.007	0.614 ±0.001
Animal visual	0.580 ±0.004	0.567 ±0.023	0.574 ±0.012
Visual only	0.600 ±0.007	0.595 ±0.013	0.600 ±0.005
	256	512	896

E. Validation macro-F1: LSTM

Full	0.823 ±0.005	0.823 ±0.001	0.822 ±0.008
No group visual	0.816 ±0.007	0.813 ±0.002	0.825 ±0.014
Pose geometry	0.769 ±0.010	0.776 ±0.008	0.789 ±0.013
No pairwise geometry	0.774 ±0.008	0.783 ±0.018	0.786 ±0.007
Animal visual	0.745 ±0.018	0.740 ±0.011	0.745 ±0.006
Visual only	0.742 ±0.002	0.756 ±0.006	0.745 ±0.012
	256	512	896

F. Validation macro-F1: attention

Full	0.836 ±0.003	0.830 ±0.009	0.838 ±0.005
No group visual	0.830 ±0.004	0.835 ±0.006	0.832 ±0.006
Pose geometry	0.785 ±0.003	0.787 ±0.001	0.797 ±0.014
No pairwise geometry	0.791 ±0.007	0.785 ±0.002	0.797 ±0.005
Animal visual	0.762 ±0.003	0.755 ±0.009	0.762 ±0.005
Visual only	0.777 ±0.008	0.767 ±0.003	0.777 ±0.005
	256	512	896

G. Individual seeds for all stream/head/capacity configurations (each point = one model seed; diagonal is validation – held-out parity)

